

VaayuDrishti

National Weather Big Data Analytics Platform

Frontend Specification - Complete API Reference

Problem ID	SIH26069 (26069)
Organization	Ministry of Earth Sciences (MoES) / IMD
Purpose	Everything a frontend team needs to build Phase 2
Contents	32 REST endpoints + WebSocket, request/response contracts, auth model, error codes, and per-page feature mapping
Base URL	http://localhost:8000 (API prefix: /api/v1)
Interactive docs	http://localhost:8000/docs (Swagger UI)
Machine contract	http://localhost:8000/openapi.json
Verified by	backend/scripts/api_smoke.py - 49/49 checks passing
Version	1.0
Date	30 September 2026

1. Base URL, Conventions & Error Handling

1.1 Base URL and envelope conventions

REST base	http://localhost:8000/api/v1
Health base	GET /health and GET /api/v1/health (identical)
WebSocket	ws://localhost:8000/api/v1/live/stream
Content type	application/json; charset=utf-8 (except media + CSV export)
Timestamps	ISO-8601 with offset, e.g. 2026-09-30T08:30:00+00:00 (UTC)
Coordinates	WGS84 decimal degrees: lat -90..90, lon -180..180
Pagination	Every list returns {items:[...], meta:{page,page_size,total,pages}}
Unknown params	Silently ignored (they do NOT cause 422)
CORS	Enabled, default allow_origins='*' (configure via CORS_ORIGINS)

1.2 Error envelope (FastAPI default)

All errors return the standard FastAPI shape. Plan the frontend error interceptor around it:

```

{"detail": "invalid credentials"} # 401 / 403 / 404 / 409 / 429
{"detail": [{"loc": ["query", "bucket"],
              "msg": "Input should be ...",
              "type": "literal_error"}]} # 422 validation errors

```

1.3 Status-code cheat sheet

Code	When
200	Success (GET/PATCH/POST where documented)
201	Citizen report created (returns tracking token)
401	Missing/invalid/expired JWT, or bad login credentials
403	Authenticated but role too low (e.g. reviewer PATCHes sources)
404	Unknown report id, tracking token, media key, or source name
413	Citizen upload image larger than 10 MB
422	Validation error (bad enum, out-of-range, short text, bad bbox)
429	Login rate limit (15/5min/IP) or citizen limit (10/hour/IP)

2. Authentication & Session Management

2.1 Login flow

POST /api/v1/auth/login

auth: public success: 200 | 401 | 422 | 429

REQUEST

```
{"email": "admin@vaayu.local", "password": "vaayu@123"}
```

RESPONSE 200

```
{"access_token": "eyJhbGciOi...",
  "refresh_token": "eyJhbGciOi...",
  "token_type": "bearer", "role": "admin"}
```

- Store access_token (30 min TTL) in memory; refresh_token (7 day TTL) in an httpOnly-ish storage of your choice.
- Send Authorization: Bearer <access_token> on every authenticated request.
- Login rate limit: 15 attempts / 300 s per IP -> 429 {"detail": "too many login attempts, retry later"}.

2.2 Refresh flow

POST /api/v1/auth/refresh

auth: public success: 200 | 401 | 422

```
{"refresh_token": "eyJhbGciOi..."} -> TokenOut (new access + refresh pair)
```

On 401 from refresh: drop tokens and redirect to /admin/login. Refresh tokens are single-use in spirit - always persist the new pair from the response.

2.3 Current user

GET /api/v1/auth/me

auth: bearer success: 200 | 401

```
{"id": "6f1c...-uuid", "email": "admin@vaayu.local",
  "full_name": "Admin", "role": "admin"}
```

2.4 Roles

Role	Can do
admin	Everything: verification, bulk actions, source enable/interval
reviewer	Queue, verify/reject/dispute, bulk, pipeline, audit, export
viewer	Read-only admin GETs (same reads as reviewer)

Login response carries the role; gate frontend routes on it. The seeded demo admin is admin@vaayu.local / vaayu@123.

3. Data Models

3.1 Report (list item) - response of /reports, /admin/reports, search

Field	Type	Notes
id	uuid	use in /reports/{id} detail links
source	string	mastodon reddit rss imd open_meteo simulator citizen load-*
source_url	string?	original post URL
author	string?	handle/display name
text	string	post text (may contain #hashtags)
lang	string?	e.g. en, hi
hashtags	[string]?	lower-cased tags without #
lat / lon	float?	WGS84; null when ungeocoded
city / district / state	string?	gazetteer geocode
event_category	string?	one of the 8 categories (below)
severity	string?	extreme severe moderate mild unknown
confidence	float?	classifier confidence 0..1
credibility_score	float?	ML credibility 0..1
verification_status	string	pending verified rejected disputed
verification_note	string?	reviewer note
is_duplicate	bool	true = near-duplicate of duplicate_of
duplicate_of	uuid?	canonical report id
media	[MediaRef]?	see 3.3; [] or null when none
observed_at	datetime	when the event happened
ingested_at	datetime	when we stored it
plausibility	object?	weather cross-check evidence

3.2 Report detail extras - GET /reports/{id}

credibility_features object: per-feature contributions of the credibility ML model

raw object: original payload (adapter metadata, geo hints, etc.)

3.3 MediaRef

```
{
  "bucket": "vaayu-media",
  "key": "simulator/ab12....jpg",
  "content_type": "image/jpeg",
  "sha256": "...",
  "size": 48211,
  "source_url": "https://..."
}
```

Render media through GET /api/v1/reports/{id}/media/{key} (same-origin relative URL, browser gets bytes with Cache-Control: max-age=86400). Never construct MinIO URLs directly.

3.4 Fixed vocabularies

event_category rainfall, thunderstorm, flooding, heatwave, fog, dust_storm, strong_winds, other

severity extreme, severe, moderate, mild, unknown

verification_status pending, verified, rejected, disputed

media types image/jpeg, image/png, image/webp, image/gif (max 10 MB)

3.5 Envelope models

PageMeta {page, page_size, total, pages}

ReportPage {items: [Report], meta: PageMeta}

TokenOut {access_token, refresh_token, token_type, role}

CitizenSubmitOut	{id, token, tracking_path, status}
TrackOut	{id, verification_status, verification_note, observed_at, city, state, event_category}

4. Endpoint Reference

All endpoints below were exercised live by backend/scripts/api_smoke.py (49/49 checks). Parameter tables list name, type, default and constraints; anything not listed is ignored.

4.1 System

GET /health and /api/v1/health

auth: public success: 200

deep bool, default false. true -> per-component checks

```
{
  "status": "ok|degraded",
  "service": "api",
  "env": "dev",
  "components": {
    "postgres": {
      "ok": true,
      "ms": 2.1,
      "redis": {...},
      "opensearch": {
        "status": "green",
        "kafka": {...},
        "minio": {...}
      }
    }
  }
}
```

Use ?deep=true for the admin 'System Status' widget (adds ~1 s: Kafka probe runs in a worker thread).

4.2 Reports - list

GET /api/v1/reports

auth: public success: 200 | 422

Param	Type	Default	Constraints
page	int	1	>= 1
page_size	int	25	1..100
sort	enum	observed_at	observed_at ingested_at credibility_score
order	enum	desc	asc desc
date_from	str	-	ISO date/datetime (observed_at >=)
date_to	str	-	ISO date/datetime (observed_at <=)
state	str	-	exact match, e.g. Maharashtra
city	str	-	exact match
category	enum	-	one of the 8 categories
source	str	-	exact match, e.g. mastodon
verification_status	enum	-	pending verified rejected disputed
lang	str	-	e.g. en, hi
include_duplicates	bool	false	true also returns is_duplicate rows
min_credibility	float	-	0..1

Invalid sort/order value -> 422. All filters combine (AND). Duplicate rows are hidden by default - the public dashboard should keep that.

4.3 Reports - full-text search

GET /api/v1/reports/search

auth: public success: 200 | 422

Param	Type	Default	Notes
q	str	REQUIRED	1..200 chars; OpenSearch multi_match + fuzziness
page / page_size	int	1 / 25	same as list
date_from/date_to	str	-	same as list
state / category / source	str	-	same as list
include_duplicates	bool	false	same as list

Response is ReportPage. Empty q -> 422. OpenSearch down -> automatic Postgres ILIKE fallback (slower, same shape).

4.4 Reports - detail, media, tracking

GET `/api/v1/reports/{id}`

auth: public success: 200 | 404 | 422

422 when id is not a valid uuid. Returns ReportDetailOut (section 3.2).

GET `/api/v1/reports/{id}/media/{key}`

auth: public success: 200 | 404

key is the full MediaRef.key path (may contain '/'). Returns raw bytes with the stored content-type and Cache-Control: public, max-age=86400. 404 when the report, media entry, or stored object is missing.

GET `/api/v1/reports/track/{token}`

auth: public success: 200 | 404

```
{
  "id": "uuid",
  "verification_status": "pending",
  "verification_note": null,
  "observed_at": "...",
  "city": "Kolkata",
  "state": "West Bengal",
  "event_category": "flooding"
}
```

4.5 GeoJSON map layer

GET `/api/v1/geo/reports.geojson`

auth: public success: 200 | 422

Param	Type	Default	Notes
bbox	str	-	minlon,minlat,maxlon,maxlat; malformed -> 422
lat / lon	float	-	center of radius search (both required)
radius_km	float	-	>0, <=2000; used only with lat+lon
limit	int	2000	1..20000
date_from/date_to	str	-	report filters (see 4.2)
state/city/category/source	str	-	report filters
verification_status	str	-	report filter
include_duplicates	bool	false	report filter

```
{
  "type": "FeatureCollection",
  "features": [
    {
      "type": "Feature",
      "geometry": {
        "type": "Point",
        "coordinates": [77.2, 28.6]
      },
      "properties": {
        "id", "source", "city", "state", "event_category",
        "severity", "confidence", "credibility_score",
        "verification_status", "is_duplicate", "lang", "media_count",
        "observed_at", "text"
      }
    }
  ],
  "meta": {
    "count": 512,
    "limit": 2000,
    "truncated": false
  }
}
```

No coords -> feature skipped. Points always present in properties for marker styling; fit the map view to bbox for viewport-driven queries.

4.6 Analytics - KPI summary

GET `/api/v1/analytics/summary`

auth: public success: 200

No parameters. Non-duplicate reports only (all time):

```
{
  "total_reports": 11336,
  "verified": 812,
  "pending": 10480,
  "disputed": 12,
  "rejected": 32,
  "verified_pct": 7.2,
  "avg_credibility": 0.531,
  "reports_last_24h": 3904,
  "reports_last_hour": 168,
  "states_covered": 31,
  "categories": {
    "rainfall": 4021,
    "flooding": 2110,
    ...
  },
  "sources_enabled": 6,
  "computed_at": "2026-09-30T..."
}
```

4.7 Analytics - time series (charts)

GET /api/v1/analytics/timeseries

auth: public success: 200 | 422

Param	Type	Default	Notes
bucket	enum	hour	hour day week (invalid -> 422)
group_by	enum	category	category state source city lang
date_from/date_to	str	-	report filters
state / category	str	-	report filters

```
{
  "bucket": "hour",
  "group_by": "category",
  "points": [
    {
      "bucket": "2026-09-30T00:00:00+00:00",
      "key": "rainfall",
      "count": 148,
      ...
    }
  ]
}
```

Default window: last 7 days when no date filter is given. Each point row is a (time, series-key) pair - pivot by key in the chart layer.

4.8 Analytics - breakdowns

GET /api/v1/analytics/by-state

auth: public success: 200

params date_from, date_to, limit (1..50, default 36)

response {"states": [{"state": "Maharashtra", "count": 1841, "verified": 92, "avg_credibility": 0.54, "last_seen": "..."}]}

GET /api/v1/analytics/by-category

auth: public success: 200

params date_from, date_to

response {"categories": [{"category": "rainfall", "count": 4021, "avg_credibility": 0.55, "last_24h": 812, "prev_24h": 790, "trend": 22}]}

trend trend = last_24h - prev_24h (delta for up/down arrows)

GET /api/v1/analytics/by-source

auth: public success: 200

params date_from, date_to

response {"sources": [{"source": "mastodon", "count": 5012, "share_pct": 44.2, "avg_credibility": 0.51}]}

4.9 Analytics - hashtags & filter options

GET /api/v1/analytics/hashtags

auth: public success: 200

params days (1..90, default 7), limit (1..100, default 20)

response {"days": 7, "hashtags": [{"tag": "mumbai rains", "count": 143}]}

GET /api/v1/analytics/filters

auth: public success: 200

params none

response {"states": [...], "cities": [...], "sources": [...], "langs": [...], "categories": [8 values], "verification_statuses": [4 values]}

use populates the global FilterBar dropdowns on first load (cache it)

4.10 Citizen submission

POST /api/v1/citizen/reports

auth: public success: 201 | 422 | 413 | 429

Two request styles:

JSON (no file):

```
{
  "text": "Street flooding under the flyover...", # 10..4000 chars REQUIRED
  "city": "Kolkata", "state": "West Bengal",      # optional, <=120 chars
  "lat": 22.57, "lon": 88.36,                    # optional GPS
  "event_category": "flooding"                  # optional, 8-value enum
}
```

MULTIPART (with photo): same fields + file=
(image/jpeg|png|webp|gif, <= 10 MB)

201 RESPONSE

```
{
  "id": "uuid",
  "token": "G6AN29x29lOM...", # show this to the citizen
  "tracking_path": "/track/G6AN...",
  "status": "pending"
}
```

- Business rules: text trimmed to >= 10 chars; max 4 URLs; no <script>. Violations -> 422 with string detail.
- Rate limit: 10 submissions / hour / IP -> 429.
- After submit, poll GET /reports/track/{token} (section 4.4) for status.

4.11 Admin - verification queue

GET /api/v1/admin/reports

auth: bearer (reviewer+) success: 200 | 401 | 403

Param	Type	Default	Notes
page / page_size	int	1 / 25	page_size max 200 (vs 100 public)
verification_status	enum	pending	queue defaults to pending
include_duplicates	bool	true	queue shows duplicates too
date_from/date_to/state/category/source	string	-	same filters as 4.2
min_credibility	float	-	same as 4.2

Returns ReportPage (same items schema as the public list).

4.12 Admin - set verification

PATCH /api/v1/admin/reports/{id}/verification

auth: bearer (reviewer+) success: 200 | 401 | 404 | 422

```
{
  "status": "verified", # verified|rejected|disputed|pending
  "note": "confirmed by IMD bulletin" # optional, <=1000 chars
}
```

Returns the updated Report. Side effects: audit_log entry + OpenSearch reindex. 404 for unknown id.

4.13 Admin - bulk verification

POST /api/v1/admin/reports/bulk-verification

auth: bearer (reviewer+) success: 200 | 401 | 422

```
{
  "ids": ["uuid1", "uuid2", ...], # 1..200 uuids
  "status": "verified", "note": "..."}
{
  "updated": 2, "missing": ["00000000-..."], "status": "verified"
}
```

Unknown ids are reported in missing[] - the call still returns 200. ids: [] -> 422.

4.14 Admin - pipeline health

GET /api/v1/admin/pipeline

auth: bearer (reviewer+) success: 200 | 401

```
{
  "counters": {
    "sink_saved": 11336,
    "sink_duplicate": 1204,
    ...
  },
  "kafka": {
    "available": true,
    "lag": {
      "pipeline-enrich": 0,
      "pipeline-dedup": 0,
      "pipeline-classify": 0,
      "pipeline-sink": 0
    },
    "sum": 0
  },
  "stages": ["enrich", "dedup", "classify", "sink"],
  "recent_runs": [
    {
      "source": "simulator",
      "status": "ok",
      "fetched": 42,
      "published": 42,
      "errors": 0,
      "started_at": "...",
      "finished_at": "...",
      "computed_at": "..."
    }
  ]
}
```

Poll every 10 s while the admin dashboard is open; kafka.available false means the broker probe timed out (render degraded state).

4.15 Admin - ingest runs, sources, audit, export

GET /api/v1/admin/ingest-runs

auth: bearer (reviewer+) success: 200 | 401

params source (str, optional), limit (1..500, default 50)

response {"runs": [{"id", "source", "status", "started_at", "finished_at", "fetched", "published", "duplicates", "errors", "last_error"}]}

GET /api/v1/admin/sources

auth: bearer (reviewer+) success: 200 | 401

response {"sources": [{"name": "imd", "enabled": true, "interval_seconds": 600, "updated_at": "..."}]}

note array is under the 'sources' key (not a bare list)

PATCH /api/v1/admin/sources/{name}

auth: bearer (admin only) success: 200 | 401 | 403 | 404 | 422

```
{
  "enabled": false,
  "interval_seconds": 600
}
```

and/or:
5..86400

response {"name": "imd", "enabled": false, "interval_seconds": 600}

notes unknown name -> 404; reviewer role -> 403; audit-logged

GET /api/v1/admin/audit

auth: bearer (reviewer+) success: 200 | 401

params limit (1..1000, default 100), action (str filter, e.g. verification:verified)

response {"entries": [{"id", "user_id", "action", "entity", "entity_id", "before", "after", "created_at"}]}

GET /api/v1/admin/export

auth: bearer (reviewer+) success: 200 | 401 | 422

Param	Type	Default	Notes
format	enum	csv	csv json (invalid -> 422)
date_from/date_to/state/category/source	-	-	report filters
verification_status	str	-	report filter
include_duplicates	bool	false	report filter
limit	int	10000	1..50000

csv -> text/csv stream with Content-Disposition attachment vaayu_export_YYYYmmdd_HHMMSS.csv
(columns: id, source, external_id, observed_at, ingested_at, state, city, district, lat, lon, event_category,

severity, confidence, credibility_score, verification_status, is_duplicate, lang, author, source_url, text). json -> {items:[...], count:N}.

5. WebSocket Live Feed

WS **/api/v1/live/stream**

auth: public (no auth) success: 101

Server pushes Redis pub/sub messages verbatim. Three message types:

5.1 hello (first frame after connect)

```
{"type": "hello", "channel": "weather:live", "ts": "2026-09-30T..."}
```

5.2 heartbeat (every 20 s)

```
{"type": "heartbeat", "ts": "2026-09-30T..."}
```

Use it as a watchdog: if no frame arrives for > 60 s, reconnect.

5.3 report (new non-duplicate report persisted by the sink)

```
{"id": "uuid", "source": "mastodon", "city": "Pune",
  "state": "Maharashtra", "event_category": "rainfall",
  "severity": "moderate", "confidence": 0.87,
  "credibility_score": 0.62,
  "text": "Heavy rain in Pune since morning #IMD",    # <=280 chars
  "lat": 18.52, "lon": 73.86,
  "observed_at": "2026-09-30T08:31:00+00:00"}
```

- Note: report frames have no 'type' field - discriminate on presence of the 'id' field (or: type absent + id present => report).
- Client -> server frames are ignored (send pings/text freely).
- Duplicates are NOT published; rate is typically 1-3 msg/s (simulator on).
- Reconnect strategy: exponential backoff (1s, 2s, 4s ... cap 30s); on reconnect re-fetch geo/summary REST data, then apply deltas from WS.
- Same origin works: ws://localhost:8000/api/v1/live/stream (no auth header needed; cookies not used).

6. Frontend Feature Specification (Phase 2)

6.1 Recommended stack

Framework	Vite + React 18 + TypeScript
Styling	Tailwind CSS, dark/light theme toggle
Map	MapLibre GL (India boundaries, clusters, heatmap, time slider)
Charts	ECharts (timeseries, bars, donut, hashtag cloud)
Data	TanStack Query for REST + native WebSocket for live feed
State	Zustand global filter store shared by map/charts/table
Routing	React Router; /admin/* guarded by JWT role
i18n	English + Hindi dictionaries

6.2 Public pages and their endpoints

Page	Endpoints consumed
DashboardMap	geo/reports.geojson (viewport bbox), analytics/summary (KPI), analytics/timeseries (chart), analytics/filters (FilterBar), WS live/stream (new markers)
ReportsList	reports (filters+pagination+sort), analytics/filters, optional reports/search for the search box
ReportDetail	reports/{id} (detail), reports/{id}/media/{key} (gallery), geo/reports.geojson with small bbox for 'nearby'
CitizenSubmit	POST citizen/reports (JSON or multipart); show token + tracking link on success
TrackReport	reports/track/{token}; poll every 30 s while open
Analytics page	analytics/by-state, by-category, by-source, hashtags

Shared FilterBar state: date range, category[], state->city cascade, verification_status[], source[], min_credibility. Every widget reads the same store and appends its query params; /analytics/filters populates the dropdowns once per session.

6.3 Admin pages and their endpoints

Page	Endpoints consumed	Poll
Login	auth/login, auth/refresh, auth/me	-
VerificationQueue	admin/reports, PATCH admin/reports/{id}/verification, POST bulk-verification (multi-select rows)	-
PipelineHealth	admin/pipeline (lag + counters + runs)	10 s
IngestRuns	admin/ingest-runs	30 s
Sources	admin/sources GET + PATCH {name}	60 s
AuditLog	admin/audit (action filter)	60 s
Export	admin/export?format=csv (download)	-
System	health?deep=true	30 s

- Optimistic UI: apply verify/reject locally, roll back on 4xx; show the audit id in the toast for traceability.
- 403 -> banner 'reviewer role required'; 401 -> silent refresh, else redirect to login; 429 -> countdown timer from Retry-After absent, use fixed 60 s for login, 1 h for citizen submit.

6.4 Real-time behaviour

- One shared WebSocket connection at app root; fan out to map + ticker + KPI increments via a tiny event bus.

- On report frame: insert marker, bump summary counters locally, append to ticker (cap 50 rows).
- Charts refresh on a 60 s interval instead of on every WS message.

6.5 UX / i18n / accessibility

- Language toggle EN/Hi on all static copy; report text stays as-is (lang field available for badges).
- Color-code severity: extreme=red, severe=orange, moderate=yellow, mild=blue, unknown=grey; always pair color with an icon/label.
- Empty states ('no reports match filters'), skeletons for tables, error toasts with the API 'detail' string.
- CredibilityBadge: score ≥ 0.7 green, 0.4-0.7 amber, < 0.4 red; tooltip shows plausibility object when present.
- Deep-linkable URLs: keep filter state in the query string (`/[reports?state=Maharashtra&category=flooding]`).

7. Tooling & Verification

openapi.json	docs/openapi.json - exported from the live API (32 REST ops, 16 schemas)
Swagger UI	http://localhost:8000/docs - try-it-out for every endpoint
ReDoc	http://localhost:8000/redoc
Smoke script	cd backend && python -m scripts.api_smoke - 49/49 checks
Live WS test	connect to ws://localhost:8000/api/v1/live/stream and watch frames
Tests	cd backend && pytest (59 tests) ruff check . mypy .
Stack up	docker compose up -d (add --profile full for Spark)

7.1 Endpoint inventory (all 32)

PUBLIC (no auth)

GET	/health	GET	/api/v1/health
GET	/api/v1/reports	GET	/api/v1/reports/search
GET	/api/v1/reports/{id}	GET	/api/v1/reports/{id}/media/{key}
GET	/api/v1/reports/track/{token}		
GET	/api/v1/geo/reports.geojson		
GET	/api/v1/analytics/summary	GET	/api/v1/analytics/timeseries
GET	/api/v1/analytics/by-state	GET	/api/v1/analytics/by-category
GET	/api/v1/analytics/by-source	GET	/api/v1/analytics/hashtags
GET	/api/v1/analytics/filters		
POST	/api/v1/citizen/reports		
POST	/api/v1/auth/login	POST	/api/v1/auth/refresh
WS	/api/v1/live/stream		

BEARER (Authorization: Bearer ...)

GET	/api/v1/auth/me		
GET	/api/v1/admin/reports	PATCH	/api/v1/admin/reports/{id}/verification
POST	/api/v1/admin/reports/bulk-verification		
GET	/api/v1/admin/pipeline	GET	/api/v1/admin/ingest-runs
GET	/api/v1/admin/sources	PATCH	/api/v1/admin/sources/{name} (admin)
GET	/api/v1/admin/audit	GET	/api/v1/admin/export